

Apply filters to SQL queries

Project description

The project details on how to utilize SQL's basic commands and operators to create filters needed for SQL queries. Security analysts need this to analyze pieces of data from a database and determine the root of a problem based from investigations.

Retrieve after hours failed login attempts

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_time > '18:00' AND success = 0;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	astrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	astrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

19 rows in set (0.001 sec)

To discuss the results, let us define first the meaning of the command:

- SELECT means choosing the columns from the table. SELECT * means selecting all columns from the table
- FROM means which table. FROM log_in_attempts is the name of table.
- WHERE means the computing part where logic, computations, and more take place here. WHERE login_time > '18:00' AND success = 0;
 - AND operator displays record if conditions are all TRUE
 - OR operator displays record if conditions in any are TRUE
 - NOT operator displays the records where the condition is NOT TRUE.

As for the command:

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00' AND success = 0;
```

Displays the result of the SQL table where the number of failed login attempts occurred after 18:00.

Retrieve login attempts on specific dates

```

MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';

```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyawamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arussco	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1
42	ogriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
53	rwason	2022-05-08	11:51:38	CAN	192.168.133.188	1
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
58	ivelasco	2022-05-09	17:20:54	CAN	192.168.57.162	0
61	dtanaka	2022-05-09	09:45:18	USA	192.168.98.221	1
65	salonso	2022-05-09	23:42:12	MEX	192.168.52.37	1
66	aestrada	2022-05-08	21:58:32	MEX	192.168.67.223	1
67	abernard	2022-05-09	11:53:41	MEX	192.168.118.29	1
68	mrah	2022-05-08	17:16:13	US	192.168.42.248	1
70	twitchel	2022-05-09	10:55:17	MEXICO	192.168.87.199	1
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
72	alevitsk	2022-05-08	12:09:10	CANADA	192.168.139.176	1
79	abernard	2022-05-09	11:41:15	MEX	192.168.158.170	0
80	cjackson	2022-05-08	02:18:10	CANADA	192.168.33.140	1
83	lrodrigu	2022-05-08	08:10:23	USA	192.168.67.69	1
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
90	gesparza	2022-05-09	00:49:05	CANADA	192.168.87.201	0
92	pwashing	2022-05-08	00:36:12	US	192.168.247.219	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
97	jreckley	2022-05-09	02:49:23	MEXICO	192.168.32.231	1
101	sbaelish	2022-05-08	12:01:22	US	192.168.145.158	0
102	jreckley	2022-05-09	16:51:44	MEX	192.168.108.13	1
108	daquino	2022-05-09	21:30:48	CANADA	192.168.15.110	1
110	rabadi	2022-05-09	00:01:54	USA	192.168.90.124	1
112	rjensen	2022-05-09	09:22:05	MEX	192.168.69.116	1
117	bsand	2022-05-08	00:19:11	USA	192.168.197.187	0
120	twitchel	2022-05-09	02:58:17	MEXICO	192.168.134.62	0
127	abellwas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
128	jclark	2022-05-09	10:45:59	CANADA	192.168.122.169	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
134	iudhike	2022-05-09	06:46:40	USA	192.168.22.115	1
135	bsand	2022-05-09	14:06:33	US	192.168.91.238	0
144	daquino	2022-05-09	11:09:32	CANADA	192.168.139.9	0
145	ivelasco	2022-05-08	09:06:02	CANADA	192.168.39.196	1
147	yappiah	2022-05-08	06:04:34	MEX	192.168.65.245	0
148	daquino	2022-05-08	06:15:55	CANADA	192.168.135.6	1
150	rwason	2022-05-08	14:40:02	CAN	192.168.204.124	0
151	rabadi	2022-05-09	16:29:46	USA	192.168.30.225	1
158	swartell	2022-05-09	19:30:32	MEXICO	192.168.190.178	1
161	abellwas	2022-05-09	13:25:50	CAN	192.168.180.205	0
162	yappiah	2022-05-09	04:51:22	MEXICO	192.168.162.100	0
163	twitchel	2022-05-08	09:21:16	MEX	192.168.119.29	0
165	jreckley	2022-05-08	15:28:43	MEXICO	192.168.34.193	0
168	jlansky	2022-05-08	13:25:42	USA	192.168.210.94	1
169	alevitsk	2022-05-08	08:10:43	CANADA	192.168.210.228	0
170	sbaelish	2022-05-09	16:43:18	USA	192.168.65.113	0
172	rabadi	2022-05-08	08:06:50	US	192.168.180.41	1
178	sgillmore	2022-05-08	12:27:22	CAN	192.168.52.216	0
184	alevitsk	2022-05-08	03:09:48	CAN	192.168.33.70	0
186	bisles	2022-05-09	04:29:17	USA	192.168.40.72	0
187	arussco	2022-05-09	00:36:26	MEX	192.168.77.137	0
189	rwason	2022-05-08	05:37:24	CANADA	192.168.168.117	1
190	jsoto	2022-05-09	05:09:21	USA	192.168.25.60	0
191	cjackson	2022-05-08	06:46:07	CANADA	192.168.7.187	0
193	lrodrigu	2022-05-08	07:11:29	US	192.168.125.240	0
197	jsoto	2022-05-08	09:05:09	US	192.168.36.21	0

75 rows in set (0.001 sec)

Remember that OR is always TRUE if any conditions are met. Therefore, the command:

```
SELECT *
```

```
FROM log_in_attempts
```

```
WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

The results from the screenshot showed that the number of login attempts for those two days is 75.

Retrieve login attempts outside of Mexico

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT MEXICO LIKE 'MEX%';
ERROR 1054 (42S22): Unknown column 'MEXICO' in 'where clause'
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaeliah	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
21	iuduikie	2022-05-11	17:50:00	US	192.168.131.147	1
25	sbaeliah	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
38	sbaeliah	2022-05-09	14:40:01	USA	192.168.60.42	1
41	apatel	2022-05-10	17:39:42	CANADA	192.168.46.207	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
45	dtanaka	2022-05-11	10:28:54	US	192.168.223.157	1
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
161	abellmas	2022-05-09	13:25:50	CAN	192.168.180.205	0
164	jclark	2022-05-12	21:15:52	CAN	192.168.18.34	1
167	jclark	2022-05-12	15:47:45	CAN	192.168.146.51	1
168	jlansky	2022-05-08	13:25:42	USA	192.168.210.94	1
169	alevitsk	2022-05-08	08:10:43	CANADA	192.168.210.228	0
170	sbaeliah	2022-05-09	16:43:18	USA	192.168.65.113	0
171	drosas	2022-05-10	16:32:55	USA	192.168.92.218	0
172	mabadi	2022-05-08	08:06:50	US	192.168.180.41	1
173	asundara	2022-05-12	23:17:52	US	192.168.58.217	1
174	lyamamot	2022-05-10	12:26:27	US	192.168.228.122	0
175	jhill	2022-05-10	00:17:09	USA	192.168.130.218	0
177	wjaffrey	2022-05-11	00:15:55	USA	192.168.144.165	0
178	sgilmore	2022-05-08	12:27:22	CAN	192.168.52.216	0
179	jclark	2022-05-12	04:08:17	CAN	192.168.232.93	0
181	abellmas	2022-05-10	13:37:05	CAN	192.168.60.111	0
182	lyamamot	2022-05-10	06:01:31	USA	192.168.106.52	0
183	nmason	2022-05-11	05:29:36	CANADA	192.168.137.147	0
184	alevitsk	2022-05-08	03:09:48	CAN	192.168.33.70	0
185	jaoto	2022-05-10	13:34:58	USA	192.168.151.91	0
186	bisles	2022-05-09	04:29:17	USA	192.168.40.72	0
188	jaoto	2022-05-11	00:39:09	USA	192.168.21.88	0
189	nmason	2022-05-08	05:37:24	CANADA	192.168.168.117	1
190	jaoto	2022-05-09	05:09:21	USA	192.168.25.60	0
191	cjackson	2022-05-08	06:46:07	CANADA	192.168.7.187	0
192	bisles	2022-05-10	08:32:03	USA	192.168.201.40	1
193	lrodriqu	2022-05-08	07:11:29	US	192.168.125.240	0
194	jclark	2022-05-12	14:11:04	CAN	192.168.197.247	0
195	alevitsk	2022-05-11	06:59:13	CANADA	192.168.236.78	1
196	acook	2022-05-10	09:56:48	CAN	192.168.52.90	0
197	jaoto	2022-05-08	09:05:09	US	192.168.36.21	0
200	jclark	2022-05-12	01:11:45	CANADA	192.168.91.103	1

144 rows in set (0.001 sec)

For this, we utilized the pattern command 'MEX%' which meant:

- MEX% - patterns that start with MEX: MEX, MEXICO, etc.

Therefore, the command:

```
SELECT *
FROM log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

The results from the screenshot showed that FROM log_in_attempts table, selecting all columns, where the country column NOT originating from Mexico (MEX%) details 144 login attempts made outside of Mexico.

Retrieve employees in Marketing

```
MariaDB [organization]> SELECT * FROM employees WHERE department = 'Marketing' AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267
1088	k865l965m233	rgosh	Marketing	East-157
1103	NULL	randerss	Marketing	East-460
1156	a184b775c707	dellery	Marketing	East-417
1163	h679i515j339	cwilliam	Marketing	East-216

```
7 rows in set (0.001 sec)
```

The command:

```
SELECT *
```

```
FROM employees
```

```
WHERE department = 'Marketing' AND office LIKE 'East%';
```

Showed that the results of the employees table where the first row from the username column is 'elarson' from the 'Marketing' department within the 'East' offices.

Retrieve employees in Finance or Sales

```
MariaDB [organization]> SELECT * FROM employees WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgillmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuchiike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasco	Finance	East-156
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	dquinco	Finance	West-280
1047	v109x587y644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1057	f370g535h632	mscott	Sales	South-270
1062	k367l639m697	redwards	Finance	North-180
1063	l686n140o569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1069	NULL	jpark	Finance	East-110
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421
1076	y347z204a710	fgarcia	Finance	Central-270
1078	a667b270c984	sharley	Sales	North-418
1081	d647e310f618	qcorbit	Finance	South-290
1083	f840g812h544	gkoshi	Finance	West-165
1085	h339i498j269	cperez	Sales	East-325
1086	i281j129k749	lmajumda	Sales	West-499
1089	l358m929n154	jpark2	Sales	West-251
1091	n378o313p469	rtran	Sales	Central-230
1092	o391p779q935	lpark	Sales	West-227
1098	u671v146w618	tarchamb	Sales	North-423
1099	v283w690x104	anaser	Finance	West-357
1105	b551c837d758	kmei	Finance	Central-232
1107	d168e758f876	akajwara	Sales	North-471
1109	f229g533h679	nlocklea	Sales	East-196
1110	g567h376i314	pchaudnu	Sales	Central-428
1111	h835i179j862	jlee	Sales	West-309
1116	m272n572o874	nzhao	Sales	South-100
1117	n683o758p820	dahmad	Sales	West-405
1118	o305p208q337	jpark3	Sales	South-329
1119	p164q780r999	cmubarak	Sales	West-409
1121	r628s557t397	vrojas	Sales	East-288
1122	s103t952u851	btorres	Finance	West-319
1130	a317b635c465	tsnow	Sales	Central-451
1136	g299h520i457	jhawes	Finance	West-416
1138	i671j355k725	scromero	Finance	South-329
1142	m674n127o823	lsilva	Finance	East-440
1144	NULL	erobinso	Finance	Central-266
1147	r454s225t299	tvoga	Finance	West-177
1148	s328t505u907	dharvey	Finance	South-181
1159	d881e710f732	jshen	Finance	East-193
1164	i682j513k442	fmeltz	Finance	North-163
1169	NULL	mmitchel	Sales	Central-250
1174	s371t911u987	teortiz	Finance	North-428
1175	t959v687w394	jclark2	Finance	North-194
1176	u849v569x521	nliu	Sales	West-220
1181	z803a233b718	sesta	Finance	South-207
1185	d790e839f461	revens	Sales	North-330
1186	e281f433g404	sacosta	Sales	North-460
1187	f963g637h851	hbode	Finance	East-351
1188	g164h566i795	noshiro	Finance	West-252
1195	n516o853p957	orainier	Finance	East-346

71 rows in set (0.001 sec)

The command:

```
SELECT *
```

```
FROM employees
```

```
WHERE department = 'Finance' OR department = 'Sales';
```

Showed that Finance & Sales showed 71 rows. The operator used was OR since it needed 1 condition to return as TRUE. (unlike AND operator where all conditions must be met before result).

Retrieve all employees not in IT

```
MariaDB [organization]> SELECT * FROM employees WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b233e825d303	hmurano	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f343	sgillmore	Finance	South-153
1004	e218f877g708	eraab	Human Resources	South-127
1005	f551g340h964	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i859j583k571	abernard	Finance	South-170
1009	NULL	lrodrigu	Sales	South-134
1010	k242l212m542	jlanasky	Finance	South-109
1011	l748m120n401	drozas	Sales	South-282
1015	p611g262r345	jscto	Finance	North-271
1016	q793s736t208	sbaelish	Human Resources	North-229
1017	r550u824t230	jclark	Finance	North-188
1018	s310v540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430v567	arusso	Finance	West-465
1024	y976z753a267	iudulke	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrar	Marketing	West-246
1028	c603d749e374	aestrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e331f189g913	mabadi	Marketing	West-375
1031	f419j188h578	dkot	Marketing	West-408
1034	i679j565k940	hsand	Human Resources	East-484
1035	j236k303l245	hisles	Sales	South-171
1036	k650l533m205	rjensen	Marketing	Central-239
1038	m873n636o225	btang	Human Resources	Central-260
1154	y765z123a548	cbryand	Marketing	North-182
1155	z342a966b589	zjones	Human Resources	East-122
1156	a184b775c707	dellery	Marketing	East-417
1157	b264c773d977	lstein	Human Resources	Central-343
1158	c406d877e950	hnaser	Human Resources	Central-243
1159	d881e710f732	jshen	Finance	East-193
1160	e127f591g924	spham	Marketing	West-353
1163	h679i515j339	cwilliam	Marketing	East-216
1164	i682j513k442	fsmeltz	Finance	North-163
1165	j171k893l832	rwords	Marketing	South-128
1166	k495l234m708	nyoung	Marketing	Central-397
1167	l738m922n515	tblackwe	Marketing	North-443
1169	NULL	mmitchel	Sales	Central-250
1170	o156p302q359	lalvarez	Human Resources	North-278
1172	q372r826s628	alkhan	Marketing	Central-360
1173	r537s849t690	ialcazar	Marketing	South-429
1174	s371t911u987	eorliz	Finance	North-428
1175	t359u687v394	jclark2	Finance	North-194
1176	u849v569w521	nliu	Sales	West-220
1177	v691w183x928	aezra	Human Resources	East-190
1178	w986x187y885	nlannist	Marketing	North-196
1179	x174y934z376	asalas	Human Resources	North-445
1180	y131z211a578	medwards	Human Resources	Central-340
1181	z803a233b718	sesta	Finance	South-207
1183	b566c710d544	lquraish	Human Resources	East-400
1184	c986d200e170	ptsosie	Human Resources	Central-247
1185	d790e839f461	revens	Sales	North-330
1186	e281f433g404	sacosta	Sales	North-460
1187	f963g637h851	hbode	Finance	East-351
1188	g164h566i795	noshiro	Finance	West-252
1189	h784i120j837	slefkowi	Human Resources	West-342
1190	NULL	kcarter	Marketing	Central-270
1191	NULL	shakimi	Marketing	Central-366
1194	m340n287o441	zwarren	Human Resources	West-212
1195	n516o853p957	orainier	Finance	East-346
1198	q308r573s459	jwartine	Marketing	South-117
1199	r520s571t459	areyes	Human Resources	East-100

161 rows in set (0.001 sec)

The command:

```
SELECT *
FROM employees
WHERE NOT department = 'Information Technology';
```

Resulted in an output where the 'Information Technology' department is excluded from the list of departments from the employees table. The overall output is 161 rows.

Summary

The overall portfolio project utilized SQL commands to filter out contents in a table. With basic commands such as SELECT, FROM, WHERE, and operators such as AND, OR, and NOT for specific filters. The end results prepared me on how to navigate and display contents.